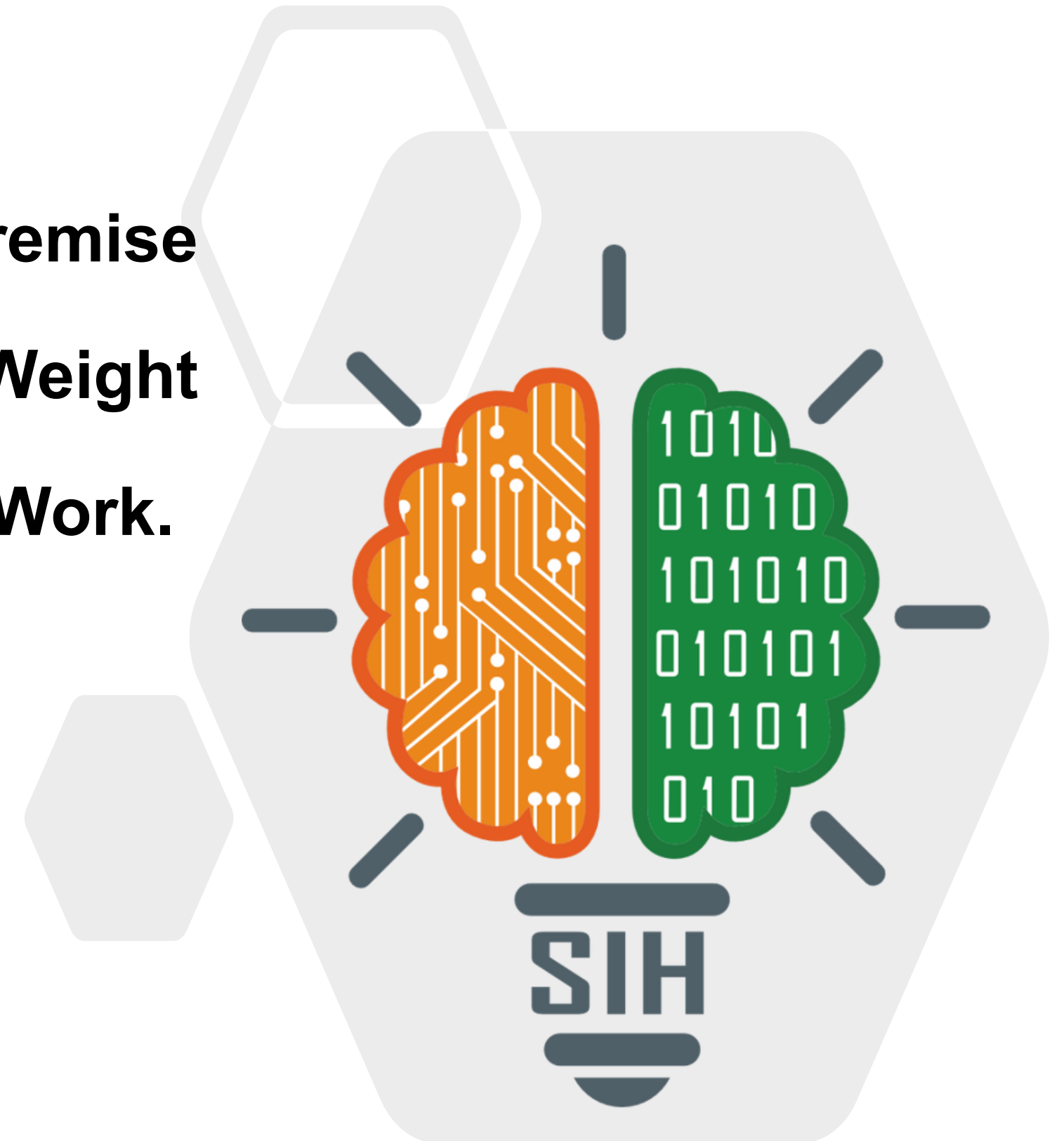


- **Problem Statement ID - SIH26117**
- **Problem Statement Title - Sovereign On-Premise Agentic AI Workbench Using Open - Weight Multimodal LLms for Confidential Industrial Work.**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID -**
- **Team Name - Tech Udaan**



1 Proposed Solution

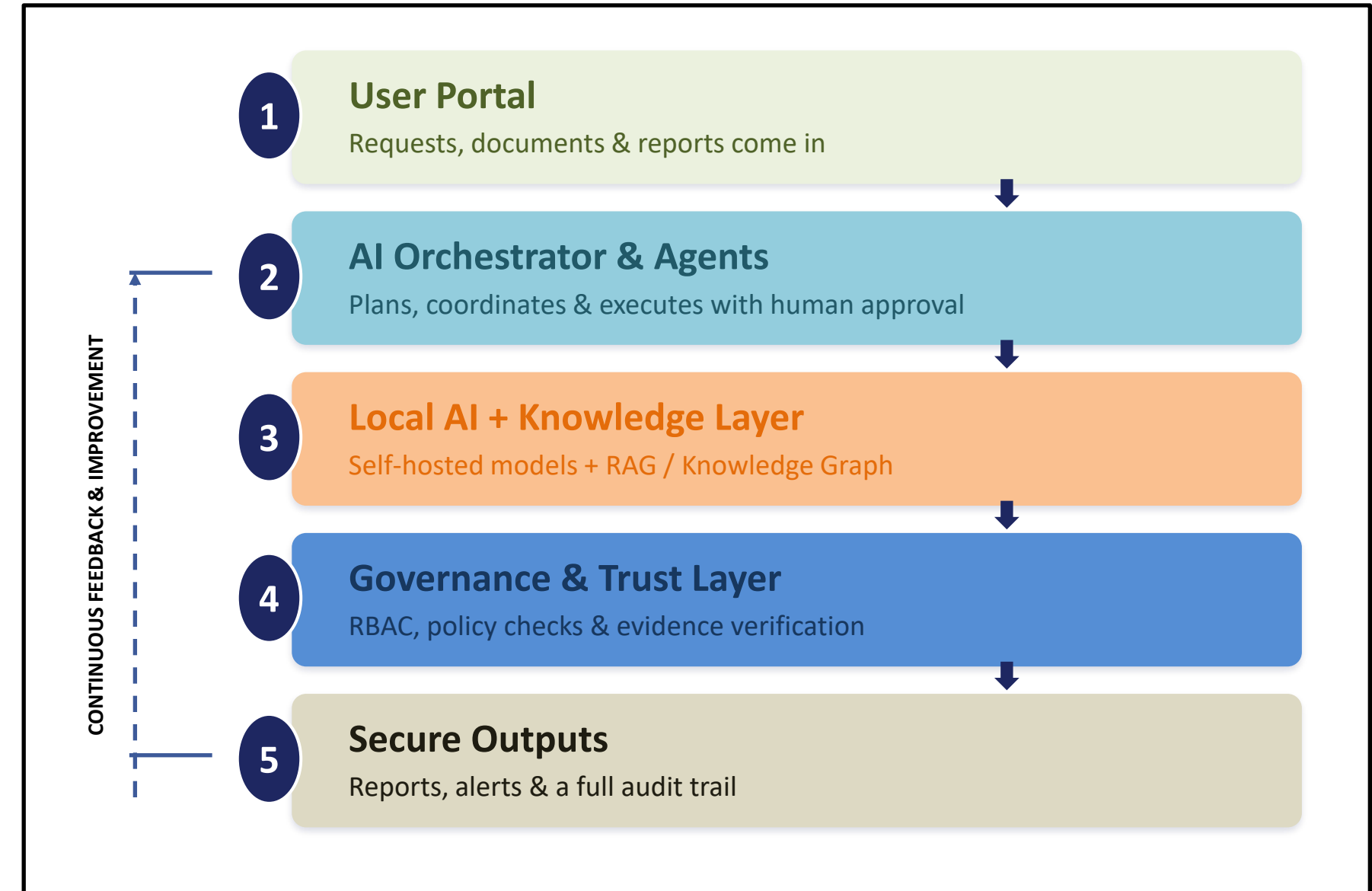
- **AI Orchestrator** — Plans tasks and manages multi-agent execution.
- **Local AI Stack** — Self-hosted LLMs, VLMs & embedding models.
- **Hybrid Knowledge Layer** — RAG + Knowledge Graphs for grounded reasoning.

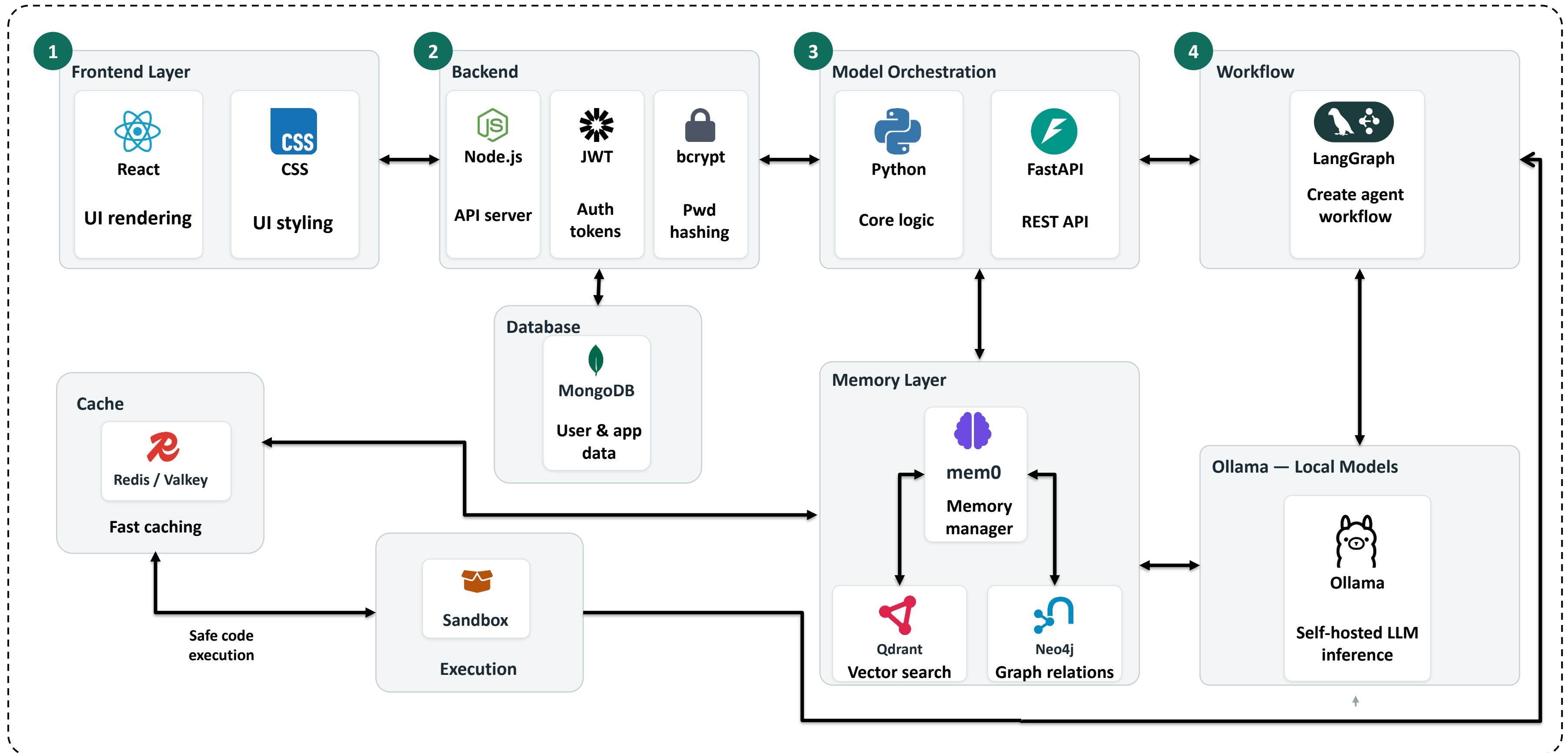
2 How It Addresses the Problem

- **Data Privacy** — Fully on-premise no external data exposure.
- **Reliable Outputs** — Evidence-based responses, hallucination-free.
- **Full Control** — RBAC, human approvals & complete audit logs.

3 Innovation & Uniqueness

- **Agentic Intelligence**
Multi-agent collaboration & RAG/Knowledge Graph, verified by an AI Trust Layer.
- **Complete & Model-Agnostic**
Full AI workbench — new open-source models plug in without redesign.
- **Governed & Auditable**
Every agent action is policy-bound and fully traceable.



 Docker — containerized services

Flow: ① Frontend → ② Backend + MongoDB → ③ Model Orchestration (pulls context from Memory Layer) → Memory Layer ↔ Ollama models → ④ Output Evaluation re-checks via Ollama → result returns to Backend → Frontend. Sandbox runs agent tool-calls via Redis/Valkey and feeds results back to Ollama.

FEASIBILITY AND VIABILITY



Feasibility

- **Proven Technology** – Multi-agent orchestration, open-weight LLMs, RAG and Graph AI are production-ready technologies.
- **On-Premise Ready** – Dockerized AI services enable secure deployment in private/air-gapped industrial environments.
- **Seamless Integration** – REST APIs, Qdrant, Neo4j and modular AI services support integration with existing enterprise systems.
- **Scalable Architecture** – Model-agnostic design allows deployment from lightweight local models to high-performance enterprise models.



Viability

- **Strong ROI** – Reduces manual analysis, operational overhead and maintenance effort through AI-assisted workflows.
- **Operational Efficiency** – Multi-agent automation accelerates document analysis, decision-making and knowledge retrieval.
- **Data Sovereignty** – Sensitive industrial data remains within the organization's infrastructure with no mandatory cloud dependency.
- **Future-Ready Platform** – Modular architecture supports new models, agents, tools and industrial use cases without redesigning the system.



Use Cases

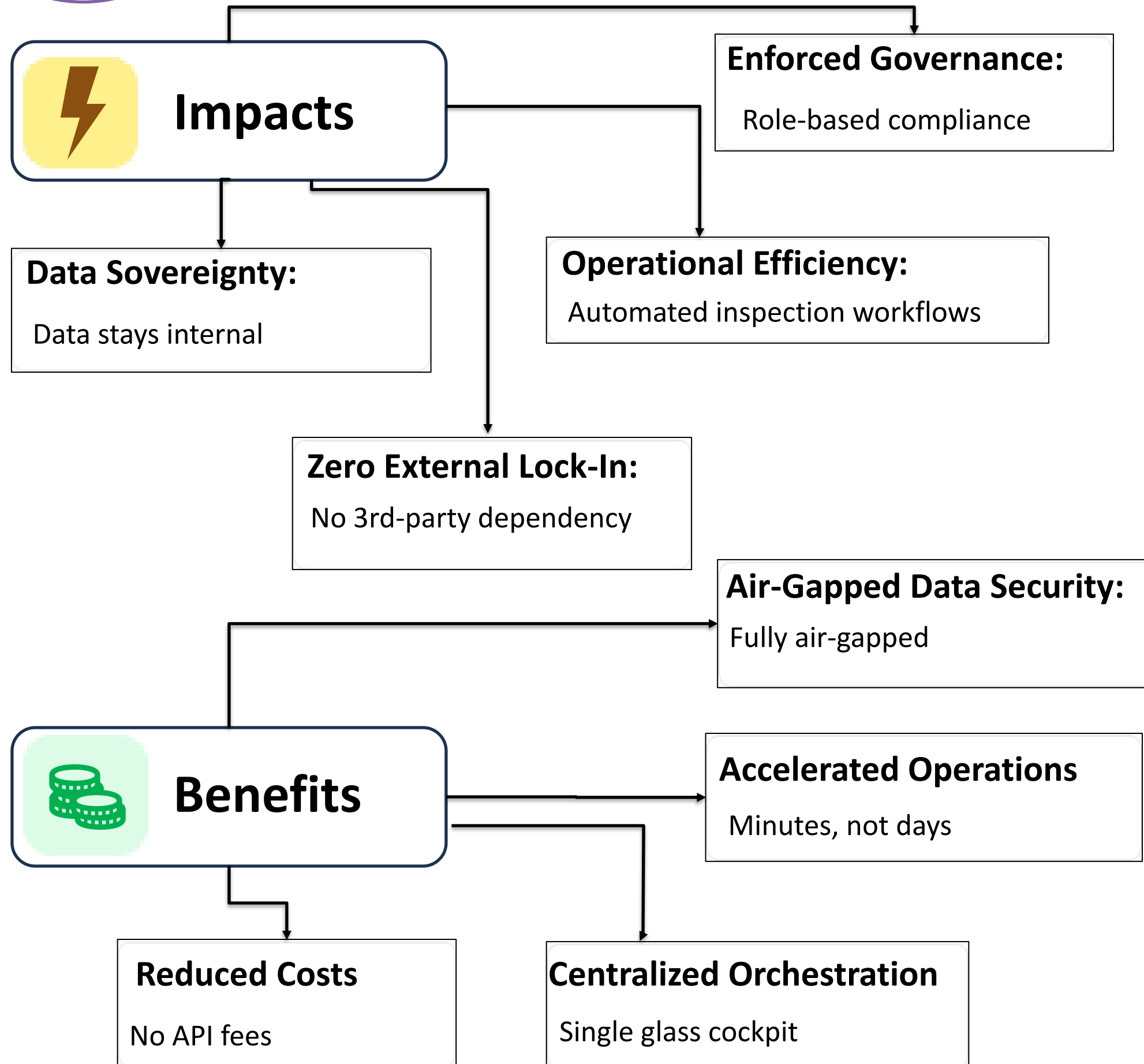
- **Intelligent Document Analysis** – Analyse SOPs, manuals, reports and engineering documents.
- **Multimodal Inspection** – Analyse equipment images, diagrams and technical documents using vision models.
- **Predictive Knowledge Assistance** – Combine RAG + Neo4j graph memory for contextual equipment insights.
- **Automated Industrial Workflows** – Multi-agent planning, analysis, verification and report generation.



Supporting Facts for Feasibility & Viability

- **Agentic AI Maturity:** Frameworks such as **LangGraph/LangChain** provide proven orchestration for multi-agent workflows, tool calling, memory and human-in-the-loop validation.
- **RAG + Graph AI: Vector databases + Neo4j Graph Memory** enable both semantic document retrieval and relationship-based reasoning across complex industrial knowledge.
- **Containerized Deployment: Docker-based microservices** allow LLM, RAG, vision, orchestration and databases to be deployed independently and scaled according to available infrastructure.
- **Industrial Applicability:** On-premise AI is particularly suitable for **oil & gas, manufacturing, power and critical infrastructure**, where confidential operational data cannot freely leave the enterprise network.

IMPACT AND BENEFITS



Challenges

- **Hardware Constraints** – Large multimodal models require high compute and memory resources.
- **Data Quality** – Inconsistent/legacy industrial documents can reduce RAG and AI accuracy.
- **Integration Complexity** – Connecting multiple enterprise systems, agents and data sources requires standardized APIs.
- **Model Reliability** – AI outputs require validation, monitoring and human oversight for critical industrial decisions.



Solutions

- ✓ **Model Optimization** – Use quantized open-weight models and hardware-aware model routing.
- ✓ **Data Pipeline** – Automated preprocessing, validation, chunking and metadata enrichment before indexing.
- ✓ **Modular Integration** – Dockerized microservices with REST APIs and standardized interfaces.
- ✓ **Human-in-the-Loop** – Agent verification, source citations, confidence scores and audit logs for critical outputs.

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COMPARISION TABLE

Feature	Existing AI Solutions	Our Sovereign AI
Model Selection	Vendor-locked models	Multi-model router
Agentic Workflow	Single-task workflows	Multi-step orchestration
Knowledge Retrieval	Cloud/internet-based	Local RAG (Qdrant)
Graph Memory	Not available	Neo4j graph memory
Multimodal Analysis	Mostly text-only	Vision + OCR + P&ID
Tool Execution	Limited tooling	Secure sandbox tools
Output Generation	Text/chat only	Reports, code, diagrams
Vendor Dependency	High vendor lock-in	Open-weight, self-hosted
Data Leaving Org	Persistent egress risk	Zero data egress
Data Privacy	Cloud-processed data	Fully on-premise